

# Haochi Wu

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## About Me

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I am Haochi Wu, a junior but passionate researcher working on low-carbon energy system for the future! I am currently a doctoral researcher in the CSS at University of Michigan (UMich), working with Prof. Michael Craig. Also I am the PhD student at Zhejiang University, and my supervisor is Prof. Mingyang Sun at Peking University.

I'm interested in renewable transition of energy system. Also I'm curious about how future energy system mitigate and adapt to climate change. Please feel free to contact me if you are interested in our research!

## EDUCATION

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**Zhejiang University**, College of Control Science and Engineering Sept 2023 – May 2025

- Ph.D. Candidate in Control Science and Engineering at ZJU;

Supervisor: Prof. [Mingyang Sun](#), Research Professor at Peking University, Honorary Lecturer at Imperial College London

- Visiting doctoral researcher at University of Michigan, Ann Arbor. UM-Center for Sustainable Systems.

Supervisor: Prof. [Michael Craig](#), Assistant Professor at University of Michigan

**Zhejiang University**, College of Control Science and Engineering Sept 2020 – May 2023

M.Sc in Control Science and Engineering

recommend as an exam-free student to the graduate school

**North China Electric Power University**, School of Control and Computer Engineering Sept 2016 – May 2020

B.Eng. in Automation (Average grade 92.3/100, Rankings 4/123)

2019 National Scholarship for Highest Academic Distinction, Ministry of Education of China

## RESEARCH INTEREST

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Power System Optimization, Energy Conversion, Macro Energy Systems, AI for Energy system

## RESEARCH PROJECT

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Contributor: sub research project modeling and manuscript writing Sept 2023 – May 2025

NSF CAREER: CAS- Climate: PI: Prof. Michael Craig

University of Michigan

Making Decarbonization of the Electric Power Sector Robust to Climate Change

Contributor: sub research project modeling and manuscript writing

Sept 2020 – May 2023

NSFC joint program with Netherlands: PI: Prof. Mingyang Sun

Zhejiang University

Data-driven Optimization of Energy Systems from the Perspectives of Information, Physics and Society

## Journal & Conference paper

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- **H. Wu**, D. Qiu, L. Zhang, M. Sun, "Adaptive Multi-Agent Reinforcement Learning for Flexible Resource Management in a Virtual Power Plant with Dynamic Participating Multi-Energy Buildings", Accepted in Applied Energy, 2024

- **H. Wu**, Q. Kong, M. Huber, et al. "Climate Change Will Increase High Temperature Risks, Degradation, and Costs of Rooftop Photovoltaics Globally", in Joule, Under Review.

- **H. Wu**, J. Wang, F. Teng, et al. "Tracking Bitcoin-Induced Carbon Trajectory in China Via Refined Spatiotemporal

Assessment". in Engineering, Under Revision.

- **H. Wu**, M. Craig. "Mitigating Thermal Risk in Buildings during Weather Related Resource Adequacy Failures with Grid-Scale Energy Storage", Accepted in International Symposium on Sustainable Systems and Technology (ISSST) 2024, Baltimore, Maryland.

- **H. Wu**, M. Craig. "Climate Change Will Increase Costs of Rooftop Photovoltaics Globally", Accepted as lighting talk in Macro Energy System Workshop (MES) 2024, Princeton University, NJ.

- L. Zhang, **H. Wu**, X. Wan, et al. "OMDA: Offline Model-Guided Distribution-Aware Offline-to-Online Reinforcement Learning". in NeurIPS 2024, Under Review.

- Y. Quan, **H. Wu**, M. Sun, et al. "PrivLoad: Privacy-preserving Load Profiles Synthesis Based on Diffusion Models". in preparation to IEEE Transactions on Smart Grid.

- J. Meng, **H. Wu**, T. Wang, et al. (2021). A novel super-cooling enhancement method for a two-stage thermoelectric cooler using integrated triangular-square current pulses. Energy, 217, 119360.

= T. Wang, **H. Wu**, J. Meng, et al. (2020). Optimization of a double-layered microchannel heat sink with semi-porous-ribs by multi-objective genetic algorithm. International Journal of Heat and Mass Transfer 149, 119217.

- J. Meng, **H. Wu**, T. Wang, et al. (2020). Thermal management of a flexible controlled thermoelectric energy conversion-utilization system using a multi-objective optimization. Applied Thermal Engineering, 179, 115721.

## HONORS & AWARDS

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Outstanding Graduate Student Award, Zhejiang University 2021

Outstanding Graduate Award, North China Electric Power University 2020

National Scholarship, Ministry of Education of China (Rate: top 1%) 2019

2nd Prize in the 12th Competition of energy conservation and emission reduction 2019

## TECHNICAL SKILL & CERTIFICATION

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**Related Courses:** Scientific Programming, Energy System Modeling, Power System Economics, Convex Optimization, Machine Learning, Earth and Climate Science

**Programming:** Python, Julia, Matlab/Simulink, Git, Latex, C, C++

**English:** TOEFL (score 100/120)